THARRANIAH TAMILWANAN

Data Science Graduate | Research & Data Analytics

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PROFESSIONAL SUMMARY

First Class Honours Data Science graduate (CGPA 3.69, Multimedia University) with hands-on research experience at Merdeka Center for Opinion Research, spanning survey analysis, M&E reporting, data validation, and stakeholder reporting. Project work in LLM factuality benchmarking, NLP, and business intelligence is backed by a strong quantitative foundation in statistics, probability, and applied mathematics. Seeking graduate roles in research, data analytics, business intelligence, and quantitative analysis.

EDUCATION

BCS (Hons.) Data Science — Multimedia University (MMU), Cyberjaya *2023 – 2026*

First Class Honours | CGPA: 3.69 | Dean's List: Terms 2310, 2430, 2530 & 2610 | MUET Band 4.5

Foundation in Information Technology — Multimedia University (MMU), Cyberjaya *Aug 2022 – Jul 2023*

CGPA: 3.52

SPM: 10 A's | PT3: 9 A's — SMK (P) Sri Aman *2017 – 2022*

WORK EXPERIENCE

Research & Data Analysis Intern — Merdeka Center for Opinion Research *Jul 2025 – Oct 2025*

- Compiled and organised election datasets across 17 Sabah constituencies to support research and accurate reporting.
- Prepared and analysed 10+ Monitoring & Evaluation (M&E) reports; cleaned, validated, and statistically analysed survey data using Google Sheets and SPSS.
- Coordinated and documented Focus Group Discussions (FGDs), supporting qualitative analysis, report preparation, and stakeholder reporting.
- Developed datasets, presentation materials, and research outputs for workshops, community engagement initiatives, and policy-related projects.

FINAL YEAR PROJECT

A Comparative Study on Factuality in Large Language Models *2025 – 2026*

Python, TruthfulQA, LLM Benchmarking, RapidFuzz, SentenceTransformers, Statistical Analysis

- Awarded 1st Runner-Up — Best Project (Data Science) for the Final Year Project.
- Designed a multi-metric benchmark comparing ChatGPT-4o mini, DeepSeek-V3.2, and Gemini 3 Flash across 817 TruthfulQA questions in 38 categories using original and paraphrased prompts.
- Evaluated factual accuracy, hallucination rate, label consistency, and paraphrase sensitivity; Gemini 3 Flash reached 82.86% accuracy while ChatGPT-4o mini achieved 90.33% consistency.
- Automated response evaluation using RapidFuzz lexical matching, SentenceTransformers semantic similarity, and rule-based classification for consistent benchmarking.

SELECTED PROJECTS

Sales Performance Dashboard — Power BI *2025*

Power BI, DAX, Data Modelling, Business Intelligence

- Built an interactive star-schema dashboard with DAX measures for revenue, profit, and order KPIs, month-over-month trends, and product-category analysis.

Handwritten Digit Recognition System *Nov 2025 – Jan 2026*

Python, OpenCV, Streamlit, MNIST

- Contributed to a group classical computer-vision pipeline using Gaussian smoothing, Otsu thresholding, watershed segmentation, and template matching against MNIST templates for multi-digit recognition.

SELECTED PROJECTS — CONTINUED

Aspect-Based Sentiment Analysis of Social Networking App Reviews *2026*

Python, NLP, scikit-learn, BiLSTM, Transformers, Streamlit, Hugging Face

- Analysed 3,118 records from six apps across nine software-quality categories and compared traditional ML, BiLSTM, and transformer approaches.
- Tuned RoBERTa achieved 82.26% accuracy and 0.820 macro F1; the project also included opinion spam-risk detection and a deployed interactive interface.

Walkable Melbourne — Interactive Pedestrian Mobility Dashboard *2026*

D3.js, JavaScript, HTML/CSS, Python, Pandas, Netlify

- Contributed to a group dashboard analysing 888,425 hourly records across 60 Melbourne sensors from 2009–2022.
- Delivered eight visualisations, six slicers, KPI cards, linked filtering, map interactions, and an animated bar-chart race to support SDG 11 pedestrian-mobility insights.

Restaurant Listing and Review System *2026*

PHP, MySQL, PDO, HTML, CSS, JavaScript

- Individually developed a web application for six restaurants with search, cuisine filtering, sorting, rating aggregation, and review CRUD.
- Implemented client/server validation, PDO prepared statements, output escaping, safe ID handling, and duplicate-review prevention.

SKILLS

Research & Analytics: Quantitative & Qualitative Research, Survey Analysis, SPSS, M&E Reporting, Data Cleaning & Validation, Trend Analysis, Insights Generation, Research Methodology

AI & LLM Evaluation: ChatGPT-4o mini, Gemini 3 Flash, DeepSeek-V3.2, Claude, TruthfulQA, LLM Benchmarking, Factuality & Hallucination Evaluation, Prompt/Response Evaluation, NLP, Sentiment Analysis

Data Analysis & BI: Python, R, SQL, Pandas, NumPy, scikit-learn, Statistical Analysis, Power BI, DAX, Excel, Google Sheets, Matplotlib

Mathematics & Quantitative: Statistics, Probability, Calculus, Discrete Mathematics, Linear Algebra & Matrices

Programming, Web & Databases: C++, PHP, MySQL, HTML, CSS, JavaScript

Tools: Jupyter Notebook, Streamlit, LaTeX, Microsoft Office, Canva

Languages: English (Fluent), Malay (Fluent), Tamil (Native)

CERTIFICATIONS

- Google — Data Analytics Professional Certificate
- Microsoft — ESG and Sustainability Analytics
- IBM SkillsBuild — Data Fundamentals | AI Fundamentals: Foundations for Understanding AI
- IBM SkillsBuild — Web Development Fundamentals | Project Management Fundamentals
- ASEAN Data Science Explorers 2026 — SAP Analytics Cloud Enablement Session
- AWS Cloud Practitioner Essentials
- HP LIFE — Data Science & Analytics
- Anthropic — AI Fluency Framework & Foundations

LEADERSHIP & ACTIVITIES

- Treasurer, Computer Club (SMK) — Managed club finances
- St. John Ambulance — Certified First Aider
- BarCamp 2025 — Event Management | Blood Donation Campaign 2025 — Logistics
- Integrity & Leadership Day 2023 — Registration